

Def 1.1. Let A be a subset of $\mathbb{R}$, denoted as $A \subseteq \mathbb{R}$.

- x is a lower bound for A if $x \leq a, (\forall) a \in A$
- x is an upper bound for A if $x \geq a, (\forall) a \in A$

$$lb(A) = \{x \in \mathbb{R} \mid x \leq a, (\forall) a \in A\}$$

$$ub(A) = \{x \in \mathbb{R} \mid x \geq a, (\forall) a \in A\}$$

$$min(A) = \inf(A) \in A \quad (\text{if } \inf(A) \in A \Rightarrow \exists min(A) = \inf(A))$$

$$max(A) = \sup(A) \in A \quad (\text{if } \sup(A) \in A \Rightarrow \exists max(A) = \sup(A))$$

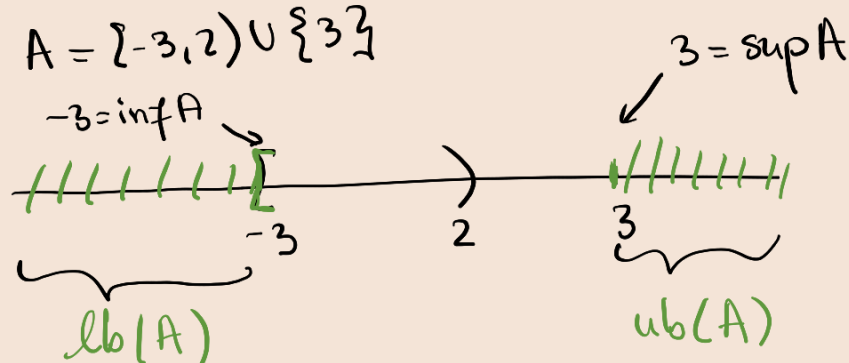
$\sup(A)$ = Smallest upper bound for A

$\inf(A)$ = largest lower bound for A

Upper bound's set: $[\sup A, +\infty)$

Lower bound's set: $(-\infty, \inf A]$

① a) $A = [-3, 2) \cup \{3\}$



$$lb(A) = (-\infty, -3]$$

$$ub(A) = [3, \infty)$$

$$min(A) = -3$$

$$max(A) = 3$$

$$\inf(A) = -3$$

$$\sup(A) = 3$$

b) $B = (-1, 1] \cup (2, \infty)$



$$\underbrace{\dots -1}_{lb(B)} \quad \dots \quad 1 \quad 2$$

$$lb(B) = (-\infty, -1]$$

$$ub(B) = \emptyset$$

$$\inf(B) = -1 \notin B \Rightarrow \nexists \min(B)$$

$$\sup(B) = \infty \notin B \Rightarrow \nexists \max(B)$$

$$x \times : A = (0, 1)$$



$$\left. \begin{array}{l} \Rightarrow \nexists \min A \\ \nexists \max A \\ \text{but} \\ \inf A = 0 \\ \sup A = 1. \end{array} \right\}$$

$$c) C = (-5, 5) \cap \mathbb{Z} = \{-4, -3, -2, \dots, 3, 4\} \text{ finite set.}$$

$$lb(C) = (-\infty, -4]$$

$$ub(C) = [4, \infty)$$

$$\inf(C) = -4 \in C \Rightarrow \exists \min C = -4.$$

$$\sup(C) = 4 \in C \Rightarrow \exists \max C = 4.$$

$$d) \emptyset$$

$$x \in lb(\emptyset) \quad \boxed{x \leq a, (\forall) a \in \emptyset} \quad P_1$$

Proof by contradiction (riducere la absurd)

Let's assume that P_1 is false.

$$\Downarrow \\ \textcolor{red}{!} P_1 \\ \Downarrow$$

$$\exists a \in \emptyset \text{ s.t. } x > a$$

$\hookrightarrow$ the empty set has nothing $\Rightarrow$ CONTRADICTION

Conclusion: P_1 true.

$$\Rightarrow lb(\emptyset) = \mathbb{R} \text{ (any real number is a lower bound for } \emptyset)$$

$\Rightarrow \text{ub}(\emptyset) = \mathbb{R}$ (any real number is a upper bound for $\emptyset$)
 $\text{inf}(\emptyset) = +\infty$ (LARGEST lower bound)
 $\text{sup}(\emptyset) = -\infty$ (SMALLEST upper bound)

② a) $A = \{x \in \mathbb{Q} \mid x^2 < 3\} = (-\sqrt{3}, \sqrt{3}) \cap \mathbb{Q}$

$\text{lb}(A) = (-\infty, -\sqrt{3}]$

$\text{ub}(A) = [\sqrt{3}, \infty)$

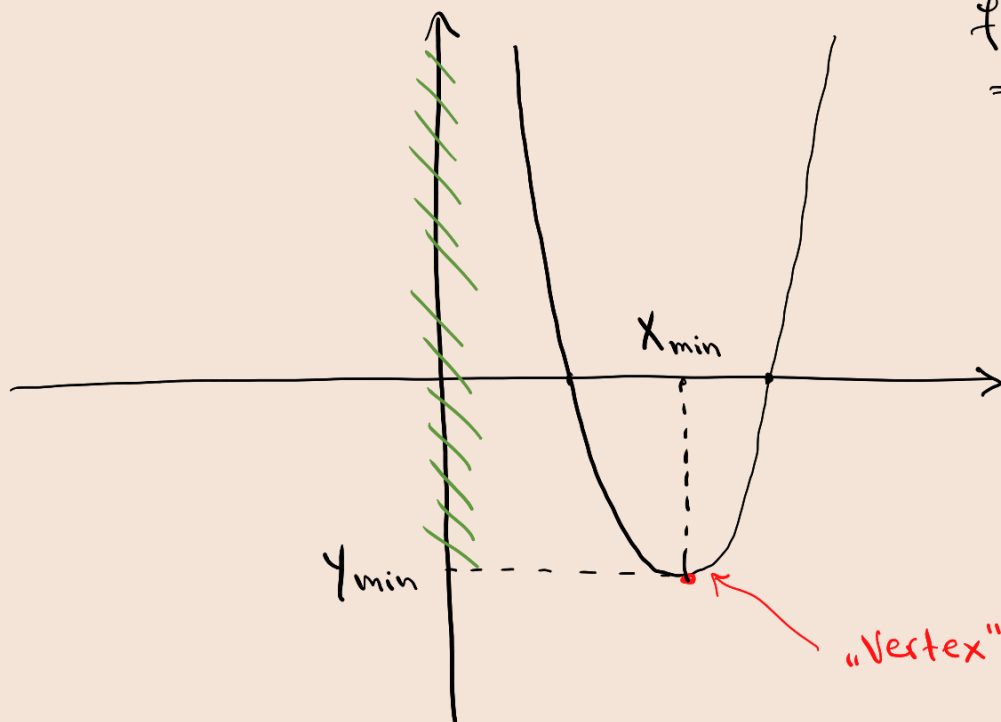
$\text{inf}(A) = -\sqrt{3} \notin A \Rightarrow \nexists \min A$

$\text{sup}(A) = \sqrt{3} \notin A \Rightarrow \nexists \max A$

b) $B = \{x^2 - 4x + 3 \mid x \in \mathbb{R}\} = \text{Im}f = \{f(x) \mid x \in \mathbb{R}\} = [-1, \infty)$

$f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = x^2 - 4x + 3 = (x-1)(x-3); a=1 > 0$

$\Downarrow$
 f has min.
 $\Rightarrow x_{\min}, y_{\min}$



$\left. \begin{matrix} x_1 = 1 \\ x_2 = 3 \end{matrix} \right\} \Rightarrow x_{\min} = 2$ (x_v is symmetric to x_1, x_2)
 $y_{\min} = -\frac{\Delta}{4a} = -1.$

$a = 1 > 0 \Rightarrow f$ has min $\Rightarrow \text{Im}f = [y_v, +\infty) = [-1, \infty)$

$\text{inf} B = -1 \in B \Rightarrow \exists \min B = -1.$

$\text{sup} B = \infty \notin B \Rightarrow \nexists \max B.$

$$c) C = \left\{ \frac{n}{n+1} \mid n \in \mathbb{N} \right\} \quad \mathbb{N} = \{0, 1, 2, \dots\}$$

$$0, \frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \dots$$

$$\inf C = 0 \in C \Rightarrow \exists \min C = 0.$$

$$\sup C = 1 \notin C \text{ (1 is the actual lim)} \Rightarrow \nexists \max C.$$

$$d) D = \left\{ 2^{-k} + 3^{-m} \mid k, m \in \mathbb{N} \right\}$$

$$\max(D) = \frac{1}{2^0} + \frac{1}{3^0} = 2 = \sup(D).$$

" k & m run over the natural numbers"

$$\inf(D) = 0 \notin D \Rightarrow \nexists \min D.$$

$$0 < \frac{1}{2^k} + \frac{1}{3^m} \in D; \quad k, m \in \mathbb{N}$$

OBS!

- A bounded set will always have inf and sup.

- Usually $\inf < \sup$, but not in $\emptyset(C)$

- Unbounded set $\Rightarrow \sup$ is ∞